



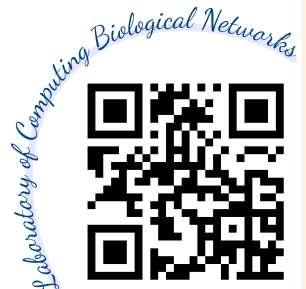
Lecture 12

2025-11-19

Optimization in neural system; Population response;
Visual receptive field; Fourier transform

- Online slides: [lec12-fourier_transform.html](#)
- Code: [code12.ipynb](#)
- Homework: [hw12.pdf](#)

Username: **cns**, Password: **nycu2025**



Information in populations of neurons

For a population response $\mathbf{r} = (r_1, r_2, r_3, \dots)$

$$H = - \int d\mathbf{r} p[\mathbf{r}] \log_2 p[\mathbf{r}] - N \log_2 \Delta r.$$

Consider response of individual neuron: $p[r_a] = \int \prod_{b \neq a} dr_b p[\mathbf{r}]$
and the entropy is $H_a = - \int d\mathbf{r} p[\mathbf{r}] \log_2 p[r_a] - \log_2 \Delta r.$

$$\sum_a H_a - H = \int d\mathbf{r} p[\mathbf{r}] \log_2 \left(\frac{p[\mathbf{r}]}{\prod_a p_a[r_a]} \right) \geq 0.$$

Thus $H \leq \sum_a H_a$. The difference is the Kullback–Leibler (KL) divergence between $p[\mathbf{r}]$ and $\prod_a p_a[r_a]$, which is a measure of **redundancy** of H from the **factorial code**.

Exact factorization & probability equalization

– maximum population response entropy

- Neurons must respond independently (factorization)
- Must have response probabilities optimal to their constraints
 - ☆ Same constraints $\Rightarrow$ Same $p[r_a]$ (equalization)

For equal response probability distribution: $\langle r_a \rangle = \langle r \rangle$,
 $\langle (r_a - \langle r \rangle)^2 \rangle = \sigma_r^2$, the covariance matrix

$$Q_{ab} = \int d\mathbf{r} p[\mathbf{r}] (r_a - \langle r \rangle) (r_b - \langle r \rangle) = \sigma_r^2 \delta_{ab}.$$

Entropy is maximized in Gaussian cases.

Decorrelation

Reduces redundancy

Variance equalization

Spreads the load of information transmission

Information and entropy maximization

— *success on visual systems*

- Explains properties of visual receptive fields in the retina, LGN, and visual cortex
- Conveying information of natural visual scenes in the presence of noise
- Requires strong assumptions about the nature of constraints relevant to the nervous system
- Analyzes only the representational properties and ignore computational goals, such as, object recognition and target tracking
- Optimizing other performance measures, different from the mutual information, may give similar results

Nevertheless, the principle of information maximization is quite successful at accounting for properties of receptive fields early in the visual pathway.

Application of information maximization

Visual receptive fields in retina, LGN, V1 for natural visual scenes in the presence of noise. Linear estimate of the response:

$$L(t) = \int_0^\infty d\tau \int d\vec{x} D(\vec{x}, \tau) s(\vec{x}, t - \tau)$$

$D(\vec{x}, \tau)$: space-time receptive field

The Separable Case

With $D(\vec{x}, \tau) = D_s(\vec{x})D_t(\tau)$ and $s(\vec{x}, t) = s_s(\vec{x})s_t(t)$, we have $L(t) = L_s L_t(t)$ where

$$L_s = \int d\vec{x} D_s(\vec{x}) s_s(\vec{x})$$
$$L_t(t) = \int_0^\infty d\tau D_t(\tau) s_t(t - \tau)$$

Spatial receptive field consideration

- Array of retinal ganglion cells covering a small patch of retina
- Spatially stationary (translational invariant)
- Cells labeled by center $\vec{a}$ of RF $D_s(\vec{x} - \vec{a})$
- Continuous limit: $\sum_{\vec{a}} \rightarrow \int d\vec{a}$

Identical and overlapping receptive field,

$$L_s(\vec{a}) = \int d\vec{x} D_s(\vec{x} - \vec{a}) s_s(\vec{x})$$

Cellular response correlation

Correlation of linear responses from two cells over all stimuli

$$Q_{LL}(\vec{a}, \vec{b}) = \left\langle L_s(\vec{a}) L_s(\vec{b}) \right\rangle \quad \text{Average over natural scenes}$$
$$= \int d\vec{x} d\vec{y} D_s(\vec{x} - \vec{a}) D_s(\vec{y} - \vec{a}) \left\langle s_s(\vec{x}) s_s(\vec{y}) \right\rangle$$

With the decorrelation and equalization assumptions:

$$Q_{LL}(\vec{a}, \vec{b}) = \sigma_L^2 \delta(\vec{a} - \vec{b}) \quad (4.37)$$

Homogeneity assumption: stimulus correlation over natural scenes depends on $|\vec{x} - \vec{y}|$ only,

$$Q_{ss}(\vec{x} - \vec{y}) = \left\langle s_s(\vec{x}) s_s(\vec{y}) \right\rangle$$

$\Rightarrow$ Want to solve for D_s .

Optimal receptive field filter

Solving for D_s using (2D) Fourier transform

$$D_s(\vec{x} - \vec{a}) = \frac{1}{4\pi^2} \int d\vec{\kappa} \exp(-i\vec{\kappa} \cdot (\vec{x} - \vec{a})) \tilde{D}_s(\vec{\kappa})$$

$$Q_{ss}(\vec{x} - \vec{y}) = \frac{1}{4\pi^2} \int d\vec{\kappa} \exp(-i\vec{\kappa} \cdot (\vec{x} - \vec{y})) \tilde{Q}_{ss}(\vec{\kappa})$$

Stimulus power spectrum $\tilde{Q}_{ss}$ is real and nonnegative.

From (4.37), the optimal RF filter satisfies

$$|\tilde{D}_s(\vec{\kappa})|^2 \tilde{Q}_{ss}(\vec{\kappa}) = \sigma_L^2$$

compensating the variation of stimulus spectrum.

Whitening filter

Eq (4.37) after FT: $\left| \tilde{D}_s(\vec{\kappa}) \right| = \frac{\sigma_L}{\sqrt{\tilde{Q}_{ss}(\vec{\kappa})}}$

Natural scenes

$$\tilde{Q}_{ss}(\vec{\kappa}) \sim 1/|\vec{\kappa}|^2$$

↪ This makes the power spectrum $\tilde{Q}_{ss}(\vec{\kappa}) \left| \tilde{D}_s(\vec{\kappa}) \right|^2$ white.

Modeling optical modulation transfer

$$\tilde{Q}_{ss}(\vec{\kappa}) \propto \frac{\exp(-\alpha |\vec{\kappa}|)}{|\vec{\kappa}|^2 + \kappa_0^2}$$

Considerations

- Excessive boosting of D at high spatial frequency κ
- Noise dominating signal at high κ not distinguished
- Full information maximization fix: need further assumptions of relevant constraints (difficult)
- Quick approximation: prefilter input to eliminate noise before maximization entropy of linear filter on the prefiltered signal

Optimization under noise

Input: $s_s(\vec{x}) + \eta(\vec{x})$ Kernel: $\tilde{D}_s(\vec{\kappa}) = \tilde{D}_w(\vec{\kappa}) \tilde{D}_\eta(\vec{\kappa})$

Spectrum:

$$\langle (s_s(\vec{x}) + \eta(\vec{x})) (s_s(\vec{y}) + \eta(\vec{y})) \rangle = Q_{ss}(\vec{x} + \vec{y}) + Q_{\eta\eta}(\vec{x} + \vec{y})$$

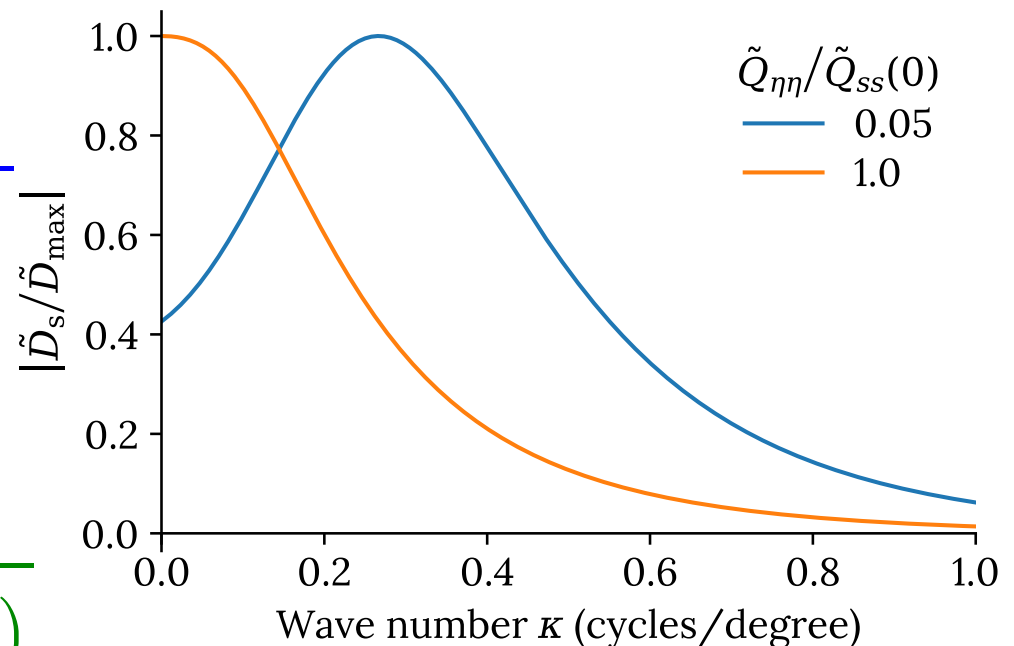
assuming no correlation between s and η .

Optimal noise suppression

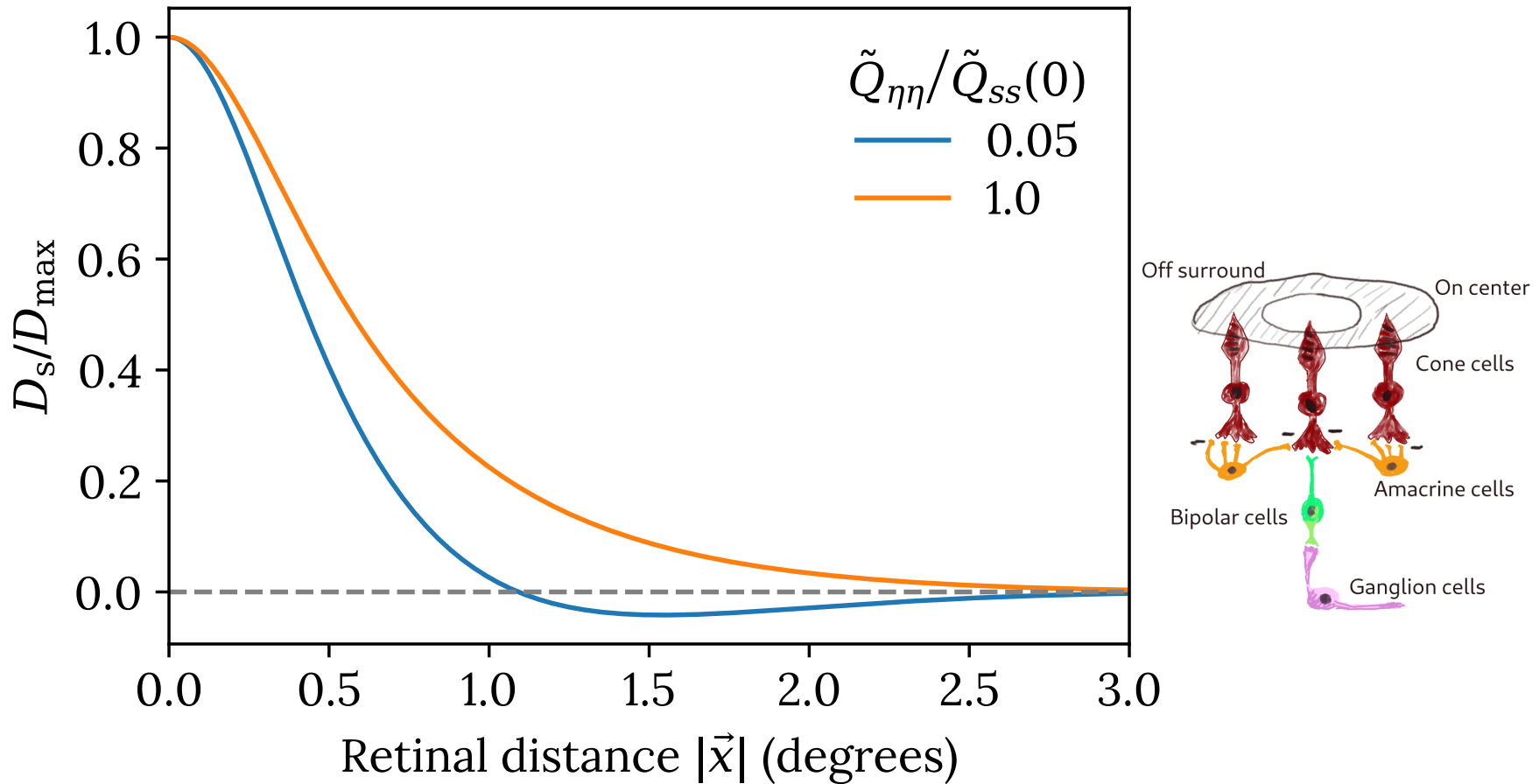
$$\tilde{D}_\eta(\vec{\kappa}) = \frac{\tilde{Q}_{ss}(\vec{\kappa})}{\tilde{Q}_{ss}(\vec{\kappa}) + \tilde{Q}_{\eta\eta}(\vec{\kappa})}$$

Using same whitening filter:

$$|\tilde{D}_s(\vec{\kappa})| \propto \frac{\sigma_L \sqrt{\tilde{Q}_{ss}(\vec{\kappa})}}{\tilde{Q}_{ss}(\vec{\kappa}) + \tilde{Q}_{\eta\eta}(\vec{\kappa})}$$



Optimal kernel in real space



Showing center-surround structure at low noise!

Parameters: $1/\alpha = 0.16$ cycles/degree, $k_0 = 0.16$ cycles/degree

Optimal assumption

Biological organisms are well “optimized” through evolution to adapt to the environment they live.

Theoretical study of biological systems:

- Target on a biological function for investigation
- Determine the functional goals and constraints involved
- Optimize for the best solutions and characteristics
- Survey experimental examples for similar solutions or characteristics

Temporal processing in the LGN

Decorrelation & Variance Equalization

$$\langle L_t(t)L_t(t') \rangle = \sigma_L^2 \delta(t - t') \quad (4.48)$$

Fourier transform to frequency space

$$D_t(\tau) = \frac{1}{2\pi} \int d\omega \exp(-i\omega\tau) \tilde{D}_t(\omega) \quad (4.49)$$

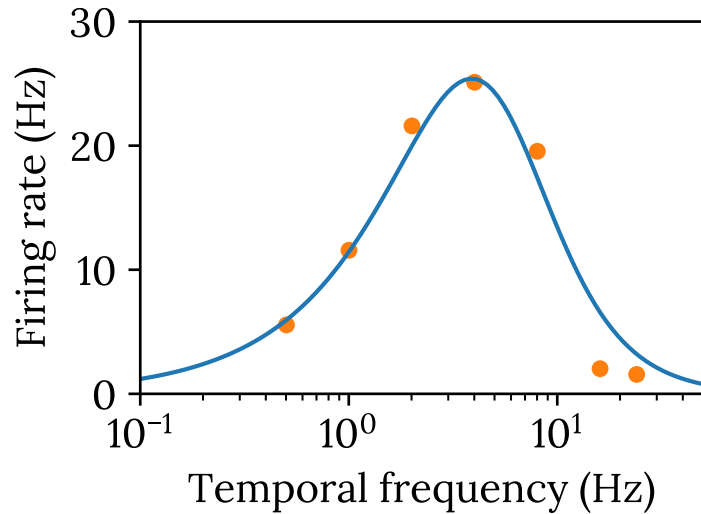
Amplitude of $\tilde{D}_t(\omega)$ is similarly given by

$$|\tilde{D}_t(\omega)| \propto \frac{\sigma_L \sqrt{\tilde{Q}_{ss}(\omega)}}{\tilde{Q}_{ss}(\omega) + \tilde{Q}_{\eta\eta}(\omega)}. \quad (4.50)$$

Temporal power spectrum of natural scenes:

$$\tilde{Q}_{ss}(\omega) \propto \frac{1}{\omega^2 + \omega_0^2} \quad \text{Dong \& Atick (1995)}$$

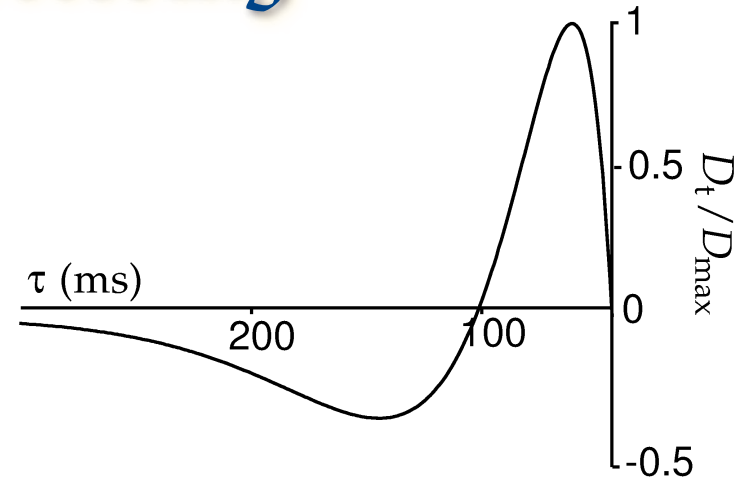
Optimal filter for temporal processing



D&A Fig. 4.4

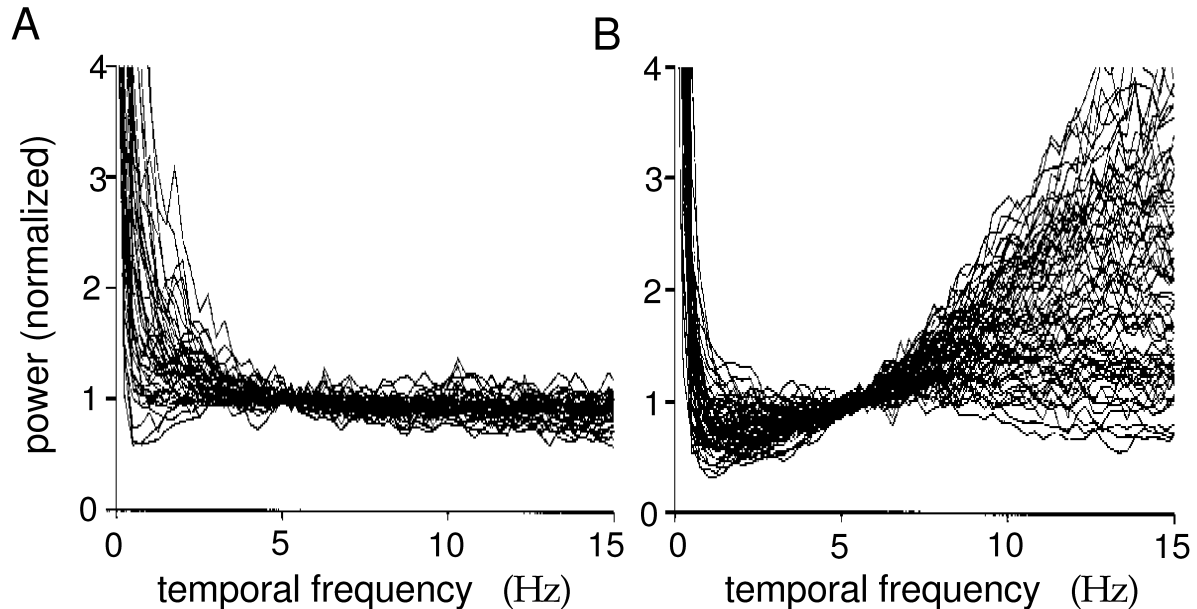
⇐ Selectivity

Causal,
minimum phase, ⇒
optimal filter



Power spectra of cat LGN cells responding to
(A) *Casablanca* the movie
(B) white-noise stimuli

D&A Fig. 4.5



Fourier transform

Following the convention of the D&A

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\kappa e^{-i\kappa x} \tilde{f}(\kappa)$$

For the inverse transform, integrate $f(x)$ to get

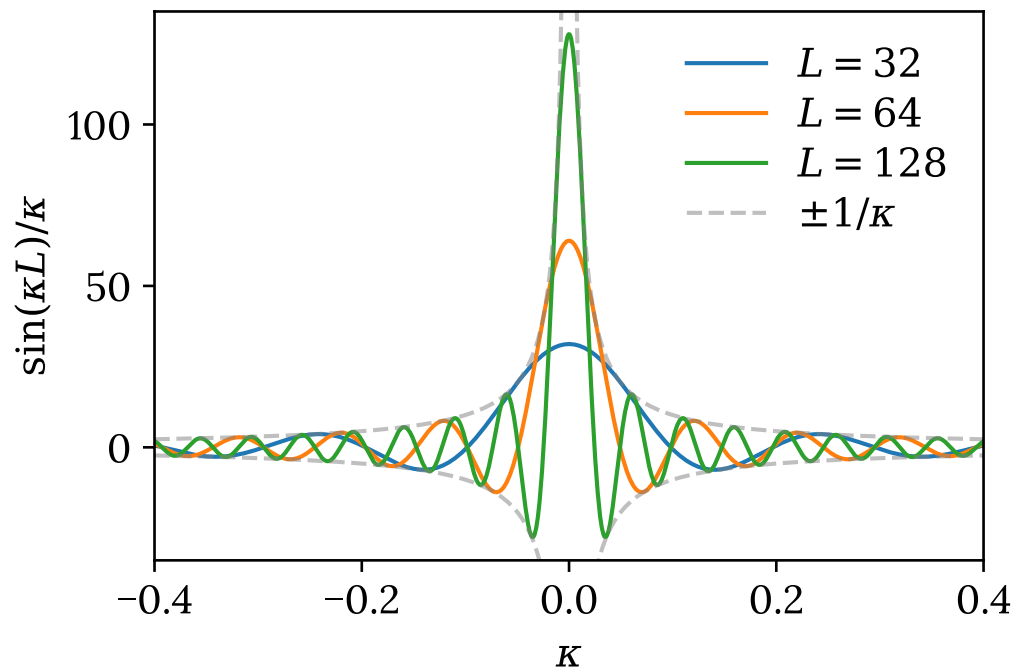
$$\begin{aligned} & \int_{-\infty}^{\infty} dx e^{i\kappa x} f(x) \\ &= \int_{-\infty}^{\infty} dx e^{i\kappa x} \frac{1}{2\pi} \int_{-\infty}^{\infty} d\kappa' e^{-i\kappa' x} \tilde{f}(\kappa') \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\kappa' \tilde{f}(\kappa') \int_{-\infty}^{\infty} dx e^{i(\kappa - \kappa')x} \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\kappa' \tilde{f}(\kappa') 2\pi \delta(\kappa - \kappa') = \tilde{f}(\kappa) \end{aligned}$$

Dirac's delta function

$$\delta(x) = 0 \text{ if } x \neq 0, \quad \int_{-\infty}^{\infty} \delta(x) dx = 1$$

Consider the function $\sin(\kappa L)/\kappa$, for $\kappa \neq 0$, it is fast oscillating for large L , thus, when convolving with a smooth function, is indistinguishable from 0.

$$\begin{aligned} & \int_{-\infty}^{\infty} dx e^{i\kappa x} \\ &= \lim_{L \rightarrow \infty} \int_{-L}^L dx e^{i\kappa x} \\ &= \lim_{L \rightarrow \infty} \frac{1}{i\kappa} (e^{i\kappa L} - e^{-i\kappa L}) \\ &= \lim_{L \rightarrow \infty} \frac{2}{\kappa} \sin(\kappa L) \end{aligned}$$



Approaching delta function

However, its improper, definite integral,

$$\int_{-\infty}^{\infty} d\kappa \frac{\sin(\kappa L)}{\kappa} = \int_{-\infty}^{\infty} d(\kappa L) \frac{\sin(\kappa L)}{\kappa L} = \pi$$

results in a finite value. Therefore, in the limit $L \rightarrow \infty$, $\sin(\kappa L)/(\pi\kappa)$ satisfies the required properties of a Dirac delta-function.

$$\delta(\kappa) = \lim_{L \rightarrow \infty} \frac{\sin(\kappa L)}{\pi\kappa}$$

Consequently, we have

$$\int_{-\infty}^{\infty} dx e^{i\kappa x} = 2\pi\delta(\kappa)$$

as used above.

Discretizing the Fourier integral

$$\begin{aligned} D(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\kappa e^{-i\kappa x} \tilde{D}(\kappa) \\ &\approx \frac{1}{2\pi} \int_{-K}^K d\kappa e^{-i\kappa x} \tilde{D}(\kappa) \\ D_m &\approx \frac{1}{2\pi} \sum_{k=0}^{N-1} \frac{2K}{N} \exp \left(-i \left(\frac{2K}{N} k - K \right) x \right) \tilde{D}_k \end{aligned}$$

$$\kappa = \frac{2K}{N} k - K, \quad x = \frac{2L}{N} m - L$$

where $k, m = 0, 1, \dots, N-1$ are integers.

Approximate CFT with DFT/FFT

$$\begin{aligned} D_m &\propto e^{iKx} \sum_{k=0}^{N-1} \exp \left[-i \frac{2K}{N} k \left(\frac{2L}{N} m - L \right) \right] \tilde{D}_k \\ &\propto \exp \frac{i2KLm}{N} \sum_{k=0}^{N-1} \exp \frac{-i4KLkm}{N^2} \exp \frac{i2KLk}{N} \tilde{D}_k \end{aligned}$$

Using $2KL = \pi N$:

$$e^{-i\pi m} D_m \propto \sum_{k=0}^{N-1} \exp \left(-2\pi i \frac{km}{N} \right) e^{i\pi k} \tilde{D}_k$$

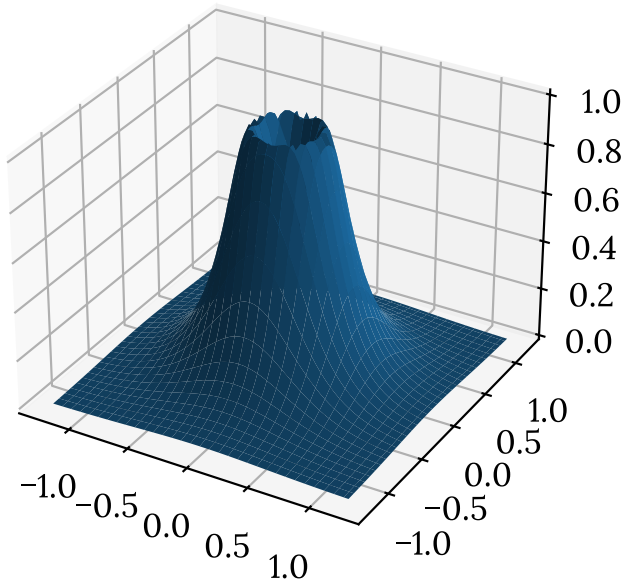
This is what FFT does in numpy!

The factors in front of D shifts the origin from the center (for CFT conventionally) to the boundary (for DFT) and vice versa.

Implementation

```
ialpha = 0.16
kappa0 = 0.16
qee = 0.05
# Power spectrum of stimulus
def Qss(k):
    return np.exp(-np.abs(k)/ialpha)/(k**2+kappa0**2)
# Optimal kernel in momentum space
def Ds_eta(k,qee):
    Q0 = Qss(0)
    Qssk = Qss(k)
    return np.sqrt(Qssk)/(qee*Q0+Qssk)
# Normalize array by its maximum
def normax(a):
    return a/np.max(a)
```

Need 2D FFT



```
from mpl_toolkits.mplot3d import Axes3D
s = normax(Ds_eta(kp,0.05)) # Normalize
w = 32 # Index range to plot
f = plt.figure()
a = f.add_subplot(projection='3d')
a.plot_surface(
    kx[N//2-w:N//2+w,N//2-w:N//2+w],
    ky[N//2-w:N//2+w,N//2-w:N//2+w],
    s[N//2-w:N//2+w,N//2-w:N//2+w]
)
plt.show()
```

```
N = 4096 # Discretization points
K = np.sqrt(np.pi*N/2) # Bounds, this makes K=L
dk = 2*K/N # Resolution in momentum (dx = dk)
rk = np.arange(N) # Enumerating K
rr = rk*2*K/N-K # All values of K
kx = np.outer(rr,np.ones(N)) # x coordinate
ky = np.outer(np.ones(N),rr) # y coordinate
# Sum of enumerations of k
kki = np.outer(rk,np.ones(N))+np.outer(np.ones(N),rk)
kp = np.sqrt(kx**2+ky**2) # Magnitude of k
```

Use the DFT in numpy to estimate CFT

- Use `fft2` from `numpy.fft` to calculate the kernel
- The phase need to be corrected by `np.exp(1j*np.pi*kk)` before and after the FFT. Or, you can use `fftshift` to shift the origin.
- Take real part when plotting...